Zhe Liu

Postdoctoral Researcher in Embodied AI and 3D Perception
Department of Computer Science, The University of Hong Kong
happinesslzcq@gmail.com | happinesslz.github.io | Google Scholar

Research Profile

I develop embodied intelligence systems that connect spatial perception, multimodal reasoning, world modeling, and executable actions. My work spans autonomous driving, 3D perception, vision-language-action models, 4D MLLMs, and world-action models, building on over eight years of research in autonomous driving and 3D perception.

40+ papers in TPAMI, TIP, CVPR, ICCV, ECCV, NeurIPS, ICML, AAAI, IROS, TITS, and ICRA. Google Scholar: **2,603 citations**; h-index **18**; i10-index **22**.

Academic Appointment and Education

2025 - Present Postdoctoral Researcher, The University of Hong Kong.
Department of Computer Science; working with Prof. Hengshuang Zhao.

2020 - 2025 Ph.D., Huazhong University of Science and Technology.
School of Electronic Information and Communications; advisor: Prof. Xiang Bai.

Research Experience

Jan 2022 - Dec 2024 Research Intern, Baidu Vision Technology Department.
Multimodal 3D object perception for autonomous driving.

Sep 2020 - Dec 2020 Research Intern, Huawei Noah's Ark Lab.
End-to-end 3D object tracking algorithms.

Research Agenda

- **Embodied data engines:** egocentric human data, human-to-robot data transformation, and corner-case feedback for scalable robotic learning.
- **VLA, world-action models, and VLN:** spatial-temporal reasoning and executable actions for manipulation, navigation, and long-horizon tasks.
- **4D world models and simulation:** physical and attention-based modeling for virtual evaluation, policy learning, and sim-to-real transfer.
- **Efficient streaming inference:** model distillation, acceleration, and streaming policies for real-time robotic deployment.
- **Intelligent robotic OS:** unified input/output interfaces and foundation models for cross-embodiment deployment.

Selected Research Projects, 2025-2026

- **AI glasses (Huawei):** efficient spatial intelligence for 3D perception, understanding, and memory retrieval on edge devices.
- **4D MLLM (Huawei):** unifying vision-action and vision-language-action capabilities in a 0.5B model.
- **4D world model (Huawei):** 4D occupancy representations for high-resolution, long-term, physically informed video generation.
- **World models and RL (Changan):** efficient attention-based world modeling for autonomous driving.
- **Streaming VLA (ACE Robotics):** memory-dependent, spatially aware vision-language-action models.
- **Ongoing:** efficient world models, efficient VLAs, and an embodied OS for physical agents.

Selected Publications | 2026

Accepted and published papers. * Equal contribution; ✉ Corresponding author; † Project leader; ‡ Equal advising.

01. **StreamPI: Streaming Multimodal Temporal Modeling for Vision-Language-Action Models**
Zhe Liu[*], Jinghua Hou[*], Yuxiang Lu, Zhenya Yang, Xianzhe Fan, Junwei Luo, Junyi Li, Ruihua Han, Zhi Hou, Hengshuang Zhao[✉]
NeurIPS, 2026. [Project] [Paper] [Code]

02. **Read It Back: Pretrained MLLMs Are Zero-Shot Reward Models for Text-to-Image Generation**
Runhui Huang, Qihui Zhang, **Zhe Liu**, Yu Gao, Jie Wu, Hengshuang Zhao
NeurIPS, 2026 - Spotlight. [Project] [Paper] [Code]

03. **EO-WM: A Physically Informed World Model for Probabilistic Earth Observation Forecasting**
Junwei Luo, Shuai Yuan, Zhenya Yang, Yansheng Li, **Zhe Liu**, Hengshuang Zhao
NeurIPS, 2026. [Paper] [Code]

04. **FlowR2A: Learning Reward-to-Action Distribution for Multimodal Driving Planning**
Xirui Li, **Zhe Liu**[†], Xiaoqing Ye[✉], Wenhua Han, Yifeng Pan, Junyu Han, Hengshuang Zhao[✉]
NeurIPS, 2026. [Project] [Paper] [Code]

05. **FASTER: Rethinking Real-Time Flow VLAs**
Yuxiang Lu, **Zhe Liu**[†], Xianzhe Fan, Zhenya Yang, Jinghua Hou, Junyi Li, Kaixin Ding, Hengshuang Zhao
NeurIPS, 2026. [Project] [Paper] [Code]

06. **SRL-MPC: Shape-Aware Reinforcement Learned Model Predictive Control**
Ruihua Han, Rui Gao, **Zhe Liu**, Xinyi Wang, Chang Chen, Shuai Wang, Qi Hao, Jia Pan, Hengshuang Zhao
NeurIPS, 2026. [Project] [Paper] [Code (coming soon)]

07. **AlphaGRPO: Unlocking Self-Reflective Multimodal Generation in UMMs via Decompositional Verifiable Reward**
Runhui Huang, Jie Wu, Rui Yang, **Zhe Liu**, Hengshuang Zhao
ICML, 2026. [Project] [Paper] [Code]

08. **GenieDrive: Towards Physics-Aware Driving World Model with 4D Occupancy Guided Video Generation**
Zhenya Yang, **Zhe Liu**[†], Yuxiang Lu, Liping Hou, Chenxuan Miao, Siyi Peng, Bailan Feng, Xiang Bai, Hengshuang Zhao[✉]
CVPR, 2026. [Paper] [Code]

09. **DrivePI: Spatial-aware 4D MLLM for Unified Autonomous Driving Understanding, Perception, Prediction and Planning**
Zhe Liu, Runhui Huang, Rui Yang, Siming Yan, Zining Wang, Lu Hou, Di Lin, Xiang Bai, Hengshuang Zhao[✉]
CVPR, 2026. [Project] [Paper] [Code]

10. **GeoTeacher: Geometry-Guided Semi-Supervised 3D Object Detection**
Jingyu Li, Xiaolong Zhao, **Zhe Liu**, Wenxiao Wu, Li Zhang
ICRA, 2026. [Paper] [Code]

11. **Visual Spatial Tuning**
Rui Yang, Ziyu Zhu, Yanwei Li, Jingjia Huang, Shen Yan, Siyuan Zhou, **Zhe Liu**, Xiangtai Li, Shuangye Li, Wenqian Wang, Yi Lin, Hengshuang Zhao
ECCV, 2026. [Paper]

Recent Preprints | 2026-2025

* Equal contribution; ✉ Corresponding author; † Project leader; ‡ Equal advising.

12. **Atomic-WAM: Execution-Grounded Atomic-Motion Reasoning for World Action Models**
Shijun Shi*, **Zhe Liu***, Zhi Hou, Yinan Chen, Tao Zhang, Kai Hu‡, Yong Liu, Jiangning Zhang‡
arXiv:2610.20157, 2026. [Project] [Paper] [Code]

13. **DVD: Dynamic Vector Decoding for Efficient MLLM-based Perception**
Jinghua Hou, **Zhe Liu**†, Hengshuang Zhao✉
arXiv:2610.12266, 2026. [Project] [Paper] [Code]

14. **PanoVLN: Towards Effective Panoramic Vision-and-Language Navigation**
Zhen Wang, Changpeng Wang, **Zhe Liu**, Zhangyang Qi, Yuxiang Lu, Zimo Zeng, Donglian Qi, Xi Chen
arXiv:2609.34759, 2026. [Project] [Paper] [Code]

15. **ACE-Brain-0.5: A Unified Embodied Foundational Model for Physical Agentic AI**
ACE-Brain Team: Ziyang Gong, Haoming Gu, Zehang Luo, Tianyi Zhang, Tao Tao, Yixiao Chi, **Zhe Liu**, Lingsi Zhu, Jingyuan Liu, Anke Tang, et al.
arXiv:2607.04426, 2026. [Project] [Paper] [Code]

16. **Metis: A Generalizable and Efficient World-Action Model for Autonomous Driving and Urban Navigation**
Jingyu Li*, **Zhe Liu***, Dongnan Hu, Junjie Wu, Zipei Ma, Wenxiao Wu, Chao Han, Zhihui Hao, Zhikang Liu, Kun Zhan, et al.
arXiv:2606.15869, 2026. [Paper] [Code]

17. **IR-SIM: A Lightweight Skill-Native Simulator for Navigation, Learning, and Benchmarking**
Ruihua Han, Shuai Wang, Chengyang Li, Rui Gao, Xinyi Wang, **Zhe Liu**, Guoliang Li, Yupu Lu, Qi Hao, Jia Pan, Hengshuang Zhao
arXiv:2606.08729, 2026. [Paper] [Code]

18. **MCNav: Memory-Aware Dynamic Cognitive Map for Zero-shot Goal-oriented Navigation**
Jingyu Li*, **Zhe Liu***, Wenxiao Wu, Li Zhang
arXiv:2605.19594, 2026. [Paper]

19. **ACE-Brain-0: Spatial Intelligence as a Shared Scaffold for Universal Embodiments**
Ziyang Gong*, Zehang Luo*, Anke Tang*, **Zhe Liu***, Shi Fu, Zhi Hou, Ganlin Yang, Weiyun Wang, Xiaofeng Wang, Jianbo Liu, et al.
arXiv:2603.03198, 2026. [Paper]

20. **UniLION: Towards Unified Autonomous Driving Model with Linear Group RNNs**
Zhe Liu, Jinghua Hou, Xiaoqing Ye, Jingdong Wang, Hengshuang Zhao✉, Xiang Bai✉
arXiv:2511.01768, 2025. [Project] [Paper] [Code]

Industry Collaborations

Previous collaborations with companies and research teams: Huawei Noah's Ark Lab; Huawei Auto BU / Yinwang; Baidu; Changan Automobile; Li Auto; ByteDance Seed; ACE Robotics; and Langyi Robotics.

Selected Publications | 2025-2019

* Equal contribution; ✉ Corresponding author; † Project leader; ‡ Equal advising.

21. **Describe, Adapt and Combine: Empowering CLIP Encoders for Open-set 3D Object Retrieval**
Zhichuan Wang, Yang Zhou, **Zhe Liu**, Rui Yu, Song Bai, Yulong Wang, Xinwei He, Xiang Bai
ICCV, 2025. [Paper] [Code]

22. **HybridTM: Combining Transformer and Mamba for 3D Semantic Segmentation**
Xinyu Wang, Jinghua Hou, **Zhe Liu**, Yingying Zhu
IROS, 2025. [Paper] [Code]

23. **LION: Linear Group RNN for 3D Object Detection in Point Clouds**
Zhe Liu*, Jinghua Hou*, Xingyu Wang, Xiaoqing Ye, Jingdong Wang, Hengshuang Zhao, Xiang Bai
NeurIPS, 2024. [Project] [Paper] [Code]

24. **SEED: A Simple and Effective 3D DETR in Point Clouds**
Zhe Liu*, Jinghua Hou*, Xiaoqing Ye, Tong Wang, Jingdong Wang, Xiang Bai
ECCV, 2024. [Paper] [Code]

25. **OPEN: Object-wise Position Embedding for Multi-view 3D Object Detection**
Jinghua Hou, Tong Wang, Xiaoqing Ye, **Zhe Liu**, Shi Gong, Xiao Tan, Errui Ding, Jingdong Wang, Xiang Bai
ECCV, 2024. [Paper] [Code]

26. **Multi-Modal 3D Object Detection by Box Matching**
Zhe Liu, Xiaoqing Ye, Zhikang Zou, Xinwei He, Xiao Tan, Errui Ding, Jingdong Wang, Xiang Bai
TITS, 2024. [Paper] [Code]

27. **Query-based Temporal Fusion with Explicit Motion for 3D Object Detection**
Jinghua Hou*, **Zhe Liu***, Dingkang Liang, Zhikang Zou, Xiaoqing Ye, Xiang Bai
NeurIPS, 2023. [Paper] [Code]

28. **A Simple Vision Transformer for Weakly Semi-supervised 3D Object Detection**
Dingyuan Zhang*, Dingkang Liang*, Zhikang Zou*, Jingyu Li, Xiaoqing Ye, **Zhe Liu**, Xiao Tan, Xiang Bai
ICCV, 2023. [Paper]

29. **DDS3D: Dense Pseudo-Labels with Dynamic Threshold for Semi-Supervised 3D Object Detection**
Jingyu Li*, **Zhe Liu***, Jinghua Hou, Dingkang Liang
ICRA, 2023. [Paper] [Code]

30. **StereoDistill: Pick the Cream from LiDAR for Distilling Stereo-based 3D Object Detection**
Zhe Liu, Xiaoqing Ye, Xiao Tan, Errui Ding, Xiang Bai
AAAI, 2023. [Paper]

31. **EPNet++: Cascade Bi-directional Fusion for Multi-Modal 3D Object Detection**
Zhe Liu, Tengteng Huang, Bingling Li, Xiwu Chen, Xi Wang, Xiang Bai
TPAMI, 2022. [Paper] [Code]

32. **EPNet: Enhancing Point Features with Image Semantics for 3D Object Detection**
Tengteng Huang*, **Zhe Liu***, Xiwu Chen, Xiang Bai
ECCV, 2020. [Paper] [Code]

33. **TANet: Robust 3D Object Detection from Point Clouds with Triple Attention**
Zhe Liu, Xin Zhao, Tengteng Huang, Ruolan Hu, Yu Zhou, Xiang Bai
AAAI, 2020 - Oral presentation. [Paper] [Code]

34. **3D Object Detection Using Scale Invariant and Feature Reweighting Networks**
Xin Zhao, **Zhe Liu**✉, Ruolan Hu, Kaiqi Huang
AAAI, 2019. [Paper]

Academic Service and Other Activities

Academic Service

Conference Area Chair: ICLR 2027 (invited in early September 2026); **ICLR 2026**.

Conference Reviewer: CVPR, ICCV, ECCV, ICLR, ICML, NeurIPS, AAAI, IJCAI, ICRA, ICASSP, and ACM Multimedia Asia.

Journal Reviewer: IEEE TPAMI, IEEE TIP, IEEE TCSVT, IEEE TITS, IEEE RA-L, Science China Information Sciences (SCIS), and IEEE TGRS.

Invited Talks

- **Machine Intelligence (Jiqizhixin), 2020:** TANet.
- **Midea Research Institute, 2020:** 3D Object Detection in Point Clouds.
- **3D CVer, 2024:** LION.
- **Shuzihuanyu, 2024:** LION and SEED.
- **The Heart of Autonomous Driving:** UniLION and DrivePI.
- **Huawei Yinwang Intelligent Technology, Nov 2025:** a two-hour talk on LION, UniLION, and DrivePI, attended by over 100 researchers.

Teaching and Community Service

- **Teaching, 2023:** delivered the Multimodal Fusion 3D Object Detection Tutorial for The Heart of Autonomous Driving (more than eight hours).
- **Volunteer, 2023:** China-Africa Text Recognition and Natural Language Processing Innovation Forum.

Honors and Awards

- **Fall 2019:** Master's National Scholarship, School of Artificial Intelligence and Automation, Huazhong University of Science and Technology.
- **Spring 2020:** NavInfo Scholarship (top 3 among 300+ candidates).

Patents

- **Three-dimensional object detection method and system based on weighted channel features of point cloud.** Xin Zhao, Kaiqi Huang, **Zhe Liu**. U.S. patent.
- **A robust three-dimensional object detection method based on ternary attention mechanism.** Xiang Bai, **Zhe Liu**, Yu Zhou, Tengteng Huang. Chinese patent; granted.
- **A three-dimensional object detection method, device and system based on DETR structure.** Xiang Bai, **Zhe Liu**, Jinghua Hou. Chinese patent application; submitted.
- **A three-dimensional object detection method, device and system.** Xiang Bai, **Zhe Liu**, Jinghua Hou, Xinyu Wang, Hengshuang Zhao. Chinese patent application; submitted.